CHAPTER 56 | OPENCLAW | AGENT FACTORY

Meet Your Personal AI Employee

MCQ Practice — Sab 15 Lessons

Roman Urdu | Easy → Hard

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Source: agentfactory.panaversity.org

60 MCQs | 9 Sections | 15 Lessons

🤖 Section 1: AI Employee Basics & Install (L1–L2) — Q1–Q8

Q1 **Easy** L1

Chatbot aur Personal AI Employee mein sabse bada farq kya hai?

A) AI Employee zyada expensive hai

B) AI Employee aapke environment mein rehta hai, 24/7 kaam karta hai, aur aapka data aapke paas rehta hai

C) Chatbot zyada smarter hai

D) AI Employee sirf mobile pe kaam karta hai

✅ **B) AI Employee aapke environment mein rehta hai**

Restaurant vs personal cook analogy: chatbot tab kaam karta hai jab aap kholte ho, AI Employee always-on daemon ki tarah 24/7 silently run karta hai. Aapka data aapke machine pe rehta hai.

Key: Always-On + Ownership

Q2 **Easy** L1

6 dimensions mein sabse important dimension kaun si hai jo ise AAPKA employee banati hai?

A) Multi-Channel

B) Proactive

C) Ownership

D) Extensible

✅ **C) Ownership**

Baaki 5 dimensions ise employee banate hain. Ownership ise AAPKA employee banata hai. Salesforce jaise enterprise tools mein bhi pehli 5 hoti hain lekin vendor control karta hai —

Ownership wahi difference hai.

Key: Ownership = Personal

Q3 Easy L1

"Proactive" dimension ka matlab kya hai?

A) Agent bahut fast jawab deta hai

B) Agent aapke type kiye bina khud kaam shuru kar leta hai — heartbeats aur cron jobs se

C) Agent bahut zyada features rakhta hai

D) Agent dobara install hone ki zaroorat nahi hoti

✅ **B) Agent khud kaam shuru kar leta hai bina aapke type kiye**

ChatGPT sirf tab kaam karta hai jab aap type karo. AI Employee heartbeats (har 30 min) aur cron jobs (exact time) se khud check karta hai — inbox, calendar, reminders — bina puche.

Key: Proactive = No Prompting

Q4 Easy L2

OpenClaw install karne ke baad wizard band karna chahiye?

A) Haan, install complete hone ke baad immediately band karo

B) Nahi — terminal band mat karo, wizard skip mat karo, read karo jo maange

C) Sirf macOS pe band karna chahiye

D) Wizard optional hai

✅ **B) Nahi — terminal band mat karo**

"Do not close your terminal. Do not skip the wizard. Read what it asks you." Install ke baad installer directly wizard mein transition karta hai.

Key: Wizard Important

Q5 Easy L2

Sabse zyada free daily requests kaun sa Google Gemini model deta hai?

A) gemini-2.5-pro (50/day)

B) gemini-2.5-flash (250/day)

C) gemini-3.1-flash-lite-preview (500/day)

D) Sab models same quota dete hain

✅ **C) gemini-3.1-flash-lite-preview (500/day)**

Learning ke liye best model — 500 requests/day, 15/minute, 1024k context. Quota khatam ho to gemini-2.5-flash pe switch karo (alag quota).

Key: Best Free Model

Q6 Medium L2

Crash Loop kyun hota hai aur kya fix hai?

A) Internet slow hone pe — speed upgrade karo

B) gateway.mode config missing hai — `openclaw config set gateway.mode local` phir restart

C) Node.js purana hai — upgrade karo

D) WhatsApp connection problem — re-scan QR code

✅ **B) gateway.mode missing — config set + restart**

Background service (LaunchAgent/systemd) register hoti hai config complete hone se pehle. Gateway start → crash → OS restart → 18 restarts 10 min mein. Log mein: "Gateway start blocked — gateway.mode not configured"

Key: Crash Loop Fix

Q7 **Medium** L2

Auth Cache Gotcha mein kya problem hoti hai?

A) Password yaad nahi rehta

B) API key rotate karne ke baad OpenClaw cached purani key use karta hai — environment variable ignore karta hai

C) Cache browser ko slow karta hai

D) Auth timeout hone pe session delete ho jata hai

✅ **B) Cached purani key — env variable ignore**

auth-profiles.json cache environment variables se priority leta hai. Fix: `rm ~/.openclaw/agents/main/agent/auth-profiles.json` → phir provider configure karo.

Key: Cache Priority Bug

Q8 **Medium** L2

Universal Setup Pattern ke 5 steps sahi order mein kaun se hain?

A) Security → Install → LLM → Channel → Verify

B) Install → LLM Configure → Channel Connect → Verify → Security

C) Channel → Install → LLM → Security → Verify

D) LLM → Install → Channel → Verify → Security

✅ **B) Install → LLM → Channel → Verify → Security**

Har agent framework ka yahi pattern hai: Runtime install → Intelligence configure → I/O connect → End-to-end verify → Security secure (pairing mode).

Key: Universal Pattern

🔧 Section 2: Agent Loop & Brain Customization (L3–L4) — Q9–Q16

Q9 **Easy** L3

Agent Loop mein Step 4 kab skip hota hai?

A) Jab internet nahi hota

B) Jab model decide kare ke koi tool zaroori nahi (knowledge se jawab ho sake)

C) Jab user fast message bheje

D) Jab tool profile "minimal" ho

✅ **B) Jab model decide kare ke tool zaroori nahi**

Task A (knowledge): Loop steps 1,2,3,5 — Step 4 skip. Dashboard mein koi tool badge nahi. Agent training data se jawab deta hai, machine ya internet touch nahi karta.

Key: Agent Loop Step 4

Q10 **Easy** L3

Dashboard mein "Exec" badge kab dikhta hai?

A) Jab agent internet search kare

B) Jab agent aapki machine pe shell command run kare

C) Jab agent memory save kare

D) Jab agent subagent spawn kare

✅ **B) Shell command run kare aapki machine pe**

Task B: "List 5 largest files in home directory" — agent khud command run karta hai aapke machine pe. Dashboard mein "Exec" badge dikhta hai aur shell command + output click karke dekh sakte ho.

Key: Exec Badge

Q11 **Medium** L3

Tool profile change karne ke liye kya zaroori hai?

A) Gateway restart zaroori hai

B) OpenClaw reinstall karna hoga

C) Kuch nahi — Dashboard → Agents → Tools → Quick Preset click karo → Save — immediately apply

D) Config file manually edit karni hogi

✅ **C) Kuch nahi — immediately apply hota hai**

Tool profile changes instant hain — gateway restart ki zaroorat nahi. Dashboard → Agents → Tools → Quick Presets mein se select karo → Save. Profile change ho jata hai.

Key: Instant Profile Change

Q12 **Easy** L4

SOUL.md file kab load hoti hai?

A) Sirf gateway restart pe

B) Din mein ek baar

C) Har session start pe sirf ek baar

D) Har message ke saath — session start pe bhi aur har message mein bhi

✅ **D) Har message ke saath**

"These five files load into every single message. Not at startup. Not once a day. Every message." SOUL.md, IDENTITY.md, AGENTS.md, TOOLS.md, USER.md — sab har message ke context mein hain.

Key: Every Message Load

Q13 **Medium** L4

Emma's Rule kya hai workspace files ke baare mein?

A) Zyada instructions likho — agent better kaam karta hai

B) Agar instruction base model se alag behavior nahi deti — delete karo (context waste hai)

C) Workspace files edit mat karo — default best hai

D) Sirf SOUL.md zaroori hai, baaki sab delete karo

✅ **B) Behavior nahi badlati — delete karo**

"Would the agent behave differently without this instruction?" — Agar jawab "nahi" hai to delete karo. Model pehle se helpful hai — "be helpful" likhna context waste hai. Sirf unique rules rakho.

Key: Emma's Rule

Q14 **Medium** L4

TOOLS.md file kya kaam karta hai?

A) Tool access grant karta hai — jo TOOLS.md mein ho agent use kar sakta hai

B) Sirf guidance notes hain — actual tool access tool profiles se control hota hai

C) MCP servers configure karta hai

D) Skills install karta hai

✅ **B) Sirf guidance notes — tool access tool profiles se**

"TOOLS.md contains guidance notes about how tools should be used. It does not grant or deny tool access." Agar tool restrict karna ho to tool profile change karo (Lesson 3), TOOLS.md se nahi.

Key: TOOLS.md Misconception

Q15 **Hard** L4

Workspace file changes kab effect karte hain?

A) Turant — next message pe

B) Gateway restart pe

C) New session pe — active conversation mein nahi

D) 24 ghante baad automatically

✅ **C) New session pe — active conversation mein nahi**

Session cache — "identity changes take effect at session boundaries, not mid-conversation." Production mein customers active conversations mein purana persona dekhte rahenge jab tak unka session expire na ho.

Key: Session Boundary

Q16 **Hard** L4

James ne chat mein "keep responses short" kaha — agent ne kal phir lambe jawab diye. Kyun?

A) Agent ne instruction ignore kiya

B) Chat mein likha tha — session memory fade hoti hai jab conversation end ho. Permanent change ke liye SOUL.md edit karna chahiye

C) Wrong model use ho raha tha

D) WhatsApp connection issue tha

✅ **B) Session memory fade hoti hai — SOUL.md edit karo**

"Session memory fades when the conversation ends. The agent's permanent personality lives in files on your machine." Chat mein kaha = temporary. Workspace file mein likha = permanent.

Key: Session vs File Memory

💾 Section 3: Memory, Commands & Skills (L5–L6) — Q17–Q24

Q17 **Easy** L5

Memory save karne ke liye message ka sahi tarika kya hai?

A) "Remember this: ..."

B) "Note karo: ..."

C) "Save in memory: ..."

D) "Please save ..."

✅ **C) "Save in memory: ..."**

"Remember this:" se agent "Got it!" bolta hai lekin file nahi likhta. "Save in memory:" actual file write trigger karta hai. Dashboard mein tool badge check karo — file write confirm hoga.

Key: Memory Save Trigger

Q18 **Easy** L5

MEMORY.md aur memory/ directory mein kya farq hai?

A) Dono same cheez hain — alag locations par

B) MEMORY.md = curated long-term (har session), memory/ = daily journals (aaj + kal auto-load)

C) memory/ har session load hoti hai, MEMORY.md sirf search pe

D) MEMORY.md delete ho jati hai har hafte

✅ **B) MEMORY.md = long-term (har session) | memory/ = daily (aaj + kal)**

MEMORY.md preferences/facts store karta hai — har session pe load hoti hai. memory/YYYY-MM-DD.md daily logs hain — aaj aur kal auto-load, purane memory_search se milte hain.

Key: Two Memory Locations

Q19 **Medium** L5

Slash command aur natural language mein kya farq hai performance ka?

A) Koi farq nahi

B) Natural language faster hoti hai

C) Slash command instant (gateway intercept, zero tokens), natural language slow (model process karta hai)

D) Slash commands sirf macOS pe kaam karte hain

✅ **C) Slash instant (zero tokens), natural language slow**

"/status" → instant, no thinking indicator, gateway intercept karta hai, model tak pahuncha hi nahi. "What model are you using?" → slow, model process karta hai, tokens spend hote hain.

Key: Slash = Gateway Level

Q20 **Medium** L5

/compact command kya karta hai?

A) Files compress karta hai

B) Gateway restart karta hai

C) Purane conversation turns ko summarize karta hai — context free karta hai jab conversation slow lage

D) Workspace files delete karta hai

✅ **C) Purane turns summarize — context free**

Jab conversation bohot lambi ho jaye aur agent slow lage, /compact older turns ko summarize kar deta hai. Gateway bhi automatically compaction karta hai jab context bhar jaye — pehle agent ko important cheezein memory mein save karne ka reminder deta hai.

Key: Context Management

Q21 **Easy** L6

ClawHub kya hai?

A) OpenClaw ka official website

B) Public skill registry — npm for agent capabilities (vector search se)

C) Agent ka dashboard

D) MCP servers ka marketplace

✅ **B) Public skill registry — npm for agent capabilities**

ClawHub: "the skill dock for sharp agents" — anyone upload kar sakta hai, version kar sakta hai (npm jaise), vector search se dhundh sakte hain. `openclaw skills search booking`

Key: ClawHub = npm for Skills

Q22 **Medium** L6

Skill install karne ke baad kya karna zaroori hai?

A) Kuch nahi — automatically available ho jati hai

B) SOUL.md update karo

C) Gateway restart karo — skills session start pe snapshot hoti hain

D) Skill manually activate karo command se

✅ **C) Gateway restart karo**

"Skills are snapshotted when a session starts. After installing a skill, restart the gateway so the next session picks it up." `openclaw gateway restart` phir `openclaw skills list` se verify karo.

Key: Restart After Install

Q23 **Medium** L6

Skill, MCP Server, aur Plugin mein kab konsa choose karna chahiye? Sahi match kaun sa hai?

A) Skill = external API access | MCP = knowledge | Plugin = gateway feature

B) Skill = knowledge/workflow | MCP = external access | Plugin = gateway capability

C) Sab ek hi kaam karte hain — preference ki baat hai

D) Plugin = external access | Skill = gateway | MCP = knowledge

✅ **B) Skill = knowledge | MCP = external | Plugin = gateway**

Escalation path: pehle Skill try karo (knowledge + workflow). Agar external live access chahiye to MCP. Agar gateway ki capability change karni ho (TTS, approval hooks) to Plugin — last resort.

Key: Escalation Path

Q24 **Hard** L6

Progressive Disclosure in SKILL.md ka kya matlab hai?

A) Skill sirf paid users ke liye available hoti hai

B) Startup pe sirf naam aur description load hoti hai (light tokens) — full instructions sirf activate hone pe

C) Skills ek ek karke manually enable karni hoti hain

D) Skill file open hone ke baad instructions dikhte hain

✅ **B) Startup pe naam + description, full instructions sirf activate pe**

Token efficiency ke liye: sirf metadata (naam, description) session start pe load hoti hai. Puri instructions tab load hoti hain jab skill trigger ho. Ye cross-platform standard hai — OpenClaw, Claude Code dono mein kaam karta hai.

Key: Token Efficient Loading

🔌 Section 4: MCP Servers & Proactive Agent (L7–L8) — Q25–Q32

Q25 **Easy** L7

MCP time server connect karne ke baad agent "What time is it in Tokyo?" ka jawab kaise deta hai?

A) Training data se — UTC+9 batata hai

B) Internet search se

C) Live tool call se — get_current_time tool use karta hai, real time milta hai

D) Aapki machine ka clock check karta hai

✅ **C) Live tool call — get_current_time**

MCP server ke baad agent generic "UTC+9" nahi deta — live specific time deta hai. Dashboard mein tool badge dikhta hai. Bina MCP ke training data se jawab hota tha.

Key: MCP = Live Access

Q26 **Medium** L7

MCP server galat package naam se configure karne ke baad agent kya karta hai?

A) Error message bhejta hai: "MCP server not found"

B) Gateway band ho jata hai

C) Silent failure — chat mein koi error nahi, training data se jawab deta hai, sirf log mein error

D) Dobara QR code maangta hai

✅ **C) Silent failure — chat mein koi error nahi**

Sabse counterintuitive behavior! Server start nahi hota → tools register nahi hote → agent jaanta hi nahi → training data se jawab. Log: `tail -20 ~/.openclaw/logs/gateway.log` mein non-zero exit code dikhega.

Key: Silent Failure

Q27 **Medium** L7

Remote MCP server (HTTP) connect karne ke liye kaunsa transport default use hota hai?

A) stdio (local process)

B) SSE (Server-Sent Events) — URL-based config ka default

C) WebSocket

D) gRPC

✅ **B) SSE — URL-based config ka default**

3 transports: stdio (local child process), SSE (remote HTTP, default for URLs), streamable-http (add "transport":"streamable-http" field). Remote: `openclaw mcp set remote '{"url":"https://..."}'`

Key: SSE Default

Q28 **Easy** L8

Heartbeat aur Cron Job mein kya farq hai?

A) Heartbeat zyada advanced hai

B) Heartbeat periodic awareness (har X min), Cron exact time delivery (exactly 9 AM)

C) Cron sirf MacOS pe kaam karta hai

D) Dono ek hi cheez hain

✅ **B) Heartbeat = periodic, Cron = exact time**

Heartbeat: "Har 30 min checklist check karo, agar kuch ho to batao, warna HEARTBEAT_OK." Cron: "Exactly 9:00 AM roz summary bhejo." Different needs ke liye alag tools.

Key: Two Proactive Mechanisms

Q29 **Medium** L8

HEARTBEAT_OK kya karta hai?

A) Agent band ho jata hai

B) Gateway ko batata hai ke system OK hai, message drop ho jata hai — phone pe kuch nahi aata

C) Memory clear ho jati hai

D) Next heartbeat schedule hota hai

✅ **B) Gateway message silently drop karta hai**

Agent checklist check karta hai → kuch attention nahi chahiye → "HEARTBEAT_OK" return karta hai → gateway ye message drop karta hai. Aapke phone pe koi message nahi aata. Ek heartbeat = one agent turn, chahe 5 checklist items hon!

Key: Silent Suppression

Q30 **Hard** L8

5 automated tasks ke liye 5 alag cron jobs vs 1 heartbeat + 1 cron — cost difference kya hai per day?

A) Koi farq nahi

B) 5 crons = 240 turns/day, 1 heartbeat + 1 cron = 50/day → 80% reduction

C) Heartbeat zyada expensive hai

D) 5 crons = 100 turns/day, heartbeat = 50/day → 50% reduction

✅ **B) 5 crons = 240 turns, heartbeat = 50 → 80% reduction**

James ka calculation: 5 crons × 2/hour × 24 hours = 240 turns. 1 heartbeat (4 items) = 48 turns + 1 cron (2 turns) = 50 turns. One heartbeat batches multiple checks into single agent turn.

Key: 80% Cost Reduction

Q31 **Medium** L8

Heartbeat routing test ke liye "target last" ka matlab kya hai?

A) Heartbeat last known agent pe jaata hai

B) Heartbeat sabse recently active conversation (last channel) pe route hota hai

C) Last user ko heartbeat jaata hai

D) Last heartbeat result repeat hota hai

✅ **B) Sabse recently active conversation pe route**

Agar aap ne pehle WhatsApp pe message kiya tha → heartbeat WhatsApp pe jayega. No phone number lookup, no typo risk. Specific number ke liye: `openclaw config set agents.defaults.heartbeat.target whatsapp`

Key: target last = Last Active

Q32 **Hard** L8

Design mein agent WHAT control karta hai aur operator WHAT control karta hai heartbeats mein?

A) Agent sab control karta hai

B) Operator sab control karta hai

C) Agent = WHAT check karna hai (HEARTBEAT.md), Operator = HOW OFTEN + WHERE (gateway config)

D) User = WHAT, Developer = HOW OFTEN

✅ **C) Agent = WHAT (HEARTBEAT.md), Operator = HOW OFTEN + WHERE (gateway config)**

Deliberate separation of concerns: Agent controls HEARTBEAT.md content (business logic). Operator controls `heartbeat.every` (frequency) aur `heartbeat.target` (delivery). Ye governance layer hai.

Key: Separation of Concerns

🎙️ Section 5: Voice, Second Agent & Google (L9–L11) — Q33–Q39

Q33 **Easy** L9

TTS production ke liye kaun sa mode recommended hai aur kyun?

A) always — sab responses voice mein

B) tagged — agent decide karta hai [[tts]] tag se

C) inbound — customer modality match karo (voice to voice, text to text)

D) off — voice ki zaroorat nahi production mein

✅ **C) inbound — modality match**

"always" annoying hai (ek word ka jawab bhi voice note), customer text copy nahi kar sakta audio se. "inbound" smart hai: customer voice bheje → voice reply, text bheje → text reply. No config needed, gateway automatically handle karta hai.

Key: Inbound = Smart Default

Q34 **Easy** L9

Microsoft Edge TTS use karne ke liye kya chahiye?

A) Paid API key

B) Microsoft account

C) Kuch nahi — free, no API key, no signup

D) Azure subscription

✅ **C) Kuch nahi — completely free**

Microsoft Edge TTS Edge browser ka "Read Aloud" backend use karta hai. No API key, no signup, no cost. 300+ neural voices. Production quality ke liye OpenAI TTS upgrade karo ($15/M chars).

Key: Free TTS Provider

Q35 **Medium** L10

Ek WhatsApp number pe 2 agents — alag alag contacts ko alag agents se kaise milwayein?

A) Dono WhatsApp numbers chahiye

B) Sender-specific routing rules — specific number pe ek agent, baaki pe doosra

C) Ye possible nahi — ek number pe ek hi agent

D) Groups se alag karo

✅ **B) Sender-specific routing rules**

`openclaw agents route --agent helper-agent --from "+923001234567"` — specific number ko helper-agent milega. Baaki sab main agent ko. Routing layer decide karta hai kaun handle kare.

Key: Sender Routing

Q36 **Medium** L10

Gateway config aur workspace config mein kya shared hai kya per-agent hai?

A) Sab kuch gateway pe share hota hai

B) Gateway = Plugins/TTS/Model (shared), Workspace = SOUL.md/Skills/MCP (per-agent)

C) Workspace = Plugins/TTS (shared), Gateway = SOUL.md (per-agent)

D) Sab workspace pe individual hota hai

✅ **B) Gateway = Plugins/TTS/Model shared | Workspace = SOUL.md/Skills/MCP per-agent**

Dono agents same tools/plugins access karte hain (gateway level). Lekin alag personalities, memories, instructions hain (workspace level). Skills ek agent ke workspace ki doosre agent ko nahi milti — intentional security design.

Key: Config Scope

Q37 **Easy** L11

Google Workspace connect karne ke liye kaunsa CLI tool use hota hai?

A) google-cli

B) gcloud

C) gog

D) google-agent

✅ **C) gog**

gog (gogcli) — "a command-line tool that bridges OpenClaw and Google Workspace." ek OAuth setup se Gmail, Calendar, Drive, Contacts, Sheets, Docs, Forms, Chat, Tasks sab access hote hain.

Key: gog CLI

Q38 **Hard** L11

Lesson 11 mein apna primary Google account kyun nahi use karna chahiye?

A) gog primary accounts support nahi karta

B) Agent ko Gmail access milne ke baad password resets, financial statements sab dikh sakta hai — security anxiety aur risk remove karne ke liye test account use karo

C) Primary account mein zyada data hota hai — slow ho jata hai

D) Google primary accounts authorize nahi karta third-party tools ko

✅ **B) Password resets, financial statements sab expose — test account use karo**

"Use a dedicated test Google account, not your primary one. This removes security anxiety and lets you experiment freely." Read-only Gmail bhi as safe nahi hai — agent sab emails dekh sakta hai including sensitive ones.

Key: Test Account First

Q39 **Medium** L11

Google Cloud Console mein "Message Content Intent" jaise setup karne ke baad kya verify karna chahiye?

A) API key email pe aata hai

B) `openclaw secrets audit` run karo — credentials safely stored hain check karo

C) Browser mein Google account check karo

D) gog --test run karo

✅ **B) openclaw secrets audit**

`openclaw secrets audit` verify karta hai ke credentials safely stored hain. Security ke liye ye important step hai especially jab sensitive Google account access set up ho.

Key: Secrets Audit

🤝 Section 6: Orchestration & Tool Gating (L12–L13) — Q40–Q44

Q40 **Medium** L12

Free-tier model sessions_spawn tool ke baare mein kya karta hai?

A) Sahi se use karta hai bina kisi problem ke

B) Tool nahi milta to error deta hai

C) Tool dekhta hai lekin hallucinated reasons de ke use nahi karta ("no permission", "limited scope")

D) Tool automatically disable ho jata hai

✅ **C) Hallucinated reasons — tool hai lekin use nahi karta**

Free-tier models "don't have permission" ya "limited to specific sessions" kehte hain — ye hallucinations hain. Platform ka koi restriction nahi. Persistent rahne ke baad finally use karta hai. Production ke liye Claude Sonnet/GPT-4 class model use karo.

Key: Model Limitation

Q41 **Hard** L12

Orchestrator pattern (subagent khud spawn kare) enable karne ke liye kya karna hoga?

A) Kuch nahi — default mein enabled hai

B) Plugin install karo

C) `openclaw config set agents.main.maxSpawnDepth 2` (default 0 hai)

D) Multi-agent plugin chahiye

✅ **C) maxSpawnDepth set karo (default 0)**

Default mein subagents khud spawn nahi kar sakte (depth 0). Coordinator pattern ke liye maxSpawnDepth badhao. Ye security by design hai — unchecked recursion prevent karta hai.

Key: maxSpawnDepth

Q42 **Medium** L13

3-Tier Safety mein Tier 2 (Plugin Hooks/requireApproval) kya karta hai?

A) Tool access completely block karta hai

B) Shell command run hone se PEHLE operator approval maangta hai — "doorbell" approach

C) API key protect karta hai

D) Sandbox mein run karta hai

✅ **B) Tool run hone se pehle approval**

James ki analogy: $500 se upar PO pe manager sign chahiye. Tool profile = door lock (open/closed). Plugin Hook = doorbell (door khula, lekin ring bajega approval ke liye). Tool hata nahi — gate lagata hai.

Key: Tier 2 = Approval Gate

Q43 **Hard** L13

Plugin build karne mein Lesson 13 ki asli skill kya hai?

A) TypeScript likhna

B) npm packages samajhna

C) Constraints likhna precisely taake agent sahi code likhe — spec writer bano, coder nahi

D) Docker images banana

✅ **C) Constraints likhna — spec writer bano**

"James ne constraints likhe, code nahi. Agent ne code likha." Lesson ka core lesson: aapke paas AI Employee hai jo software likhta hai — asli skill hai sahi spec likhna. Emma: "The actual skill: telling it what to build. Precisely enough that it does not get it wrong."

Key: Spec Writer, Not Coder

Q44 **Hard** L13

OpenClaw plugin mein silent failure kyun hoti hai?

A) Plugin naam galat ho

B) Internet band ho

C) 6 specific constraints violate hon — code compile hota hai, plugin load hota hai, lekin kuch nahi hota

D) Gateway version outdated ho

✅ **C) 6 constraints violate hon — silent failure**

"If you hand-write the code, you will hit each one." Isliye agent ko page URL dene ka pattern: agent technical constraints padhe, reference code dekhe, sahi code likhe. No error = no debugging clue — isliye constraints critical hain.

Key: Silent Plugin Failure

🚀 Section 7: Deploy & Security (L14–L15) — Q45–Q49

Q45 **Easy** L14

Beginners ke liye sabse easy deployment path kaun sa hai?

A) VPS Docker

B) VPS Native (Hetzner)

C) Managed (Alibaba Cloud) — pre-installed image, no SSH, no firewall config

D) Local machine pe hi rehne do

✅ **C) Managed (Alibaba Cloud)**

Beginners ke liye: pre-installed OpenClaw image, pre-wired Model Studio (Singapore region = 1M free tokens per model), no SSH, no firewall, no systemd. Sirf dashboard access. $0.99/month.

Key: Easiest Path

Q46 **Medium** L14

Alibaba Cloud mein Model Studio pe Singapore region kyun zaroori hai?

A) Singapore mein server fast hai

B) Singapore region mein har model pe 1 million free tokens milte hain — default region mein sab paid hai

C) Pakistan se sirf Singapore connect hota hai

D) Singapore mein Qwen Max free hai

✅ **B) Singapore = 1M free tokens per model**

Default region mein sab charged hai. Qwen Max expensive hai — zero credit account pe 24 ghante mein suspend. Singapore region select karo + free quota limit enable karo. Tabhi $0 LLM cost possible hai.

Key: Singapore = Free Tokens

Q47 **Easy** L15

NemoClaw kya hai?

A) OpenClaw ka competitor product

B) OpenClaw ka fork

C) OpenClaw + OpenShell + Privacy Router + Policy Engine — ek install command, K3s cluster

D) OpenClaw ka paid version

✅ **C) OpenClaw + OpenShell + Privacy Router + Policy Engine**

"NemoClaw is not a new product. Not a fork. Not a competitor." Ek install command → K3s (lightweight Kubernetes) cluster → 4 pods → out-of-process policy enforcement. $5/month extra.

Key: NemoClaw = 4 Components

Q48 **Hard** L15

Privacy Router kaise API key protect karta hai?

A) Key encrypt karta hai

B) Agent inference.local pe call karta hai (no key), Privacy Router alag pod mein real key add karta hai, forward karta hai — agent kabhi key nahi dekhta

C) Key weekly rotate hoti hai

D) Agent ko key nahi chahiye — model free hai

✅ **B) Agent inference.local (no key), Router alag pod mein key add karta hai**

Agent config: endpoint = "http://inference.local:8080" — no API key. Request router pod tak jaati hai (alag network namespace) → router real key add karta hai → provider forward. Agent ke paas root access bhi ho to printenv mein koi key nahi milegi.

Key: Privacy Router Architecture

Q49 **Hard** L15

NemoClaw mein upgrade kab karna chahiye?

A) Jab 1000+ users ho jayein

B) Jab server speed slow ho

C) Trust model change hone pe — customers ka data touch ho, untrusted content process ho, multi-tenant environment ho

D) Sirf enterprise companies ke liye hai

✅ **C) Trust model change hone pe — scale pe nahi**

"Evaluate when to upgrade from Docker Compose to NemoClaw based on trust model changes, not scale." Customers ka real data, untrusted prompts, multi-tenant — in cases mein upgrade. $5/month extra. Ek install command.

Key: Trust-Based Upgrade

📌 Section 9: Notes Se Missing Topics (L2,L3,L9,L10,L12,L13,L15) — Q51–Q60

Q51 **Medium** L2

OpenClaw pe message bheja lekin 30 seconds mein koi response nahi aaya — troubleshooting ke sahi steps kya hain?

A) Dobara install karo

B) `openclaw channels status --probe` → `openclaw doctor` → `tail -f ~/.openclaw/logs/gateway.log`

C) Phone restart karo aur QR code dobara scan karo

D) API key badlo aur gateway restart karo

✅ **B) channels status --probe → doctor → log**

3rd common troubleshooting problem: "No Response in 30 seconds." Step 1: channel probe karo, Step 2: doctor se health check, Step 3: gateway log mein real error dhundo. Crash Loop aur Auth Cache ke baad yeh 3rd most common installation issue hai.

Key: No Response 3-Step Fix

Q52 **Easy** L3

Dashboard mein "web_search" badge kab dikhta hai?

A) Jab agent memory save kare

B) Jab agent aapki machine pe shell command run kare

C) Jab agent internet se live search kare

D) Jab agent subagent spawn kare

✅ **C) Internet se live search kare**

Task C: "Search web for AI Agent Factory book" — agent internet se live search karta hai. Dashboard mein "web_search" badge dikhta hai. Compare 3 badges: Task A (knowledge) = koi badge nahi, Task B (machine) = "Exec" badge, Task C (internet) = "web_search" badge.

Key: web_search Badge = Internet

Q53 **Medium** L3

"messaging" tool profile mein kya disabled hota hai?

A) Sirf web search disabled hoti hai

B) File access, exec, web search — machine access sab kuch disabled

C) Sirf memory tools disabled hoti hain

D) Kuch bhi disabled nahi — "messaging" ek alias hai "full" profile ka

✅ **B) File access, exec, web search — sab disabled**

messaging profile customer-facing agents ke liye hai — sirf messaging tools available hain. File I/O, shell exec, web search sab disabled. Customer support agent ko machine pe kuch nahi karna chahiye — isliye yeh restricted profile use karo. Contrast: "full" mein sab kuch enabled hota hai.

Key: messaging = Machine Access Disabled

Q54 **Medium** L9

TTS "tagged" mode mein kaun decide karta hai ke voice note banegi ya nahi?

A) Gateway automatically decide karta hai message length dekh ke

B) Customer ki message type (voice bheje to voice, text bheje to text)

C) Agent khud — reply mein [[tts]] tag lagata hai — theory mein flexible, practice mein buggy

D) Config file mein percentage set hota hai

✅ **C) Agent [[tts]] tag lagata hai — practice mein buggy**

4 TTS modes: off (default), always (har reply voice — annoying!), inbound (modality match — production best), tagged (agent [[tts]] tag se decide kare). "tagged" theory mein flexible lagta hai lekin practice mein inconsistent hai — agent kabhi kabhi tag lagana bhool jata hai. Isliye inbound production ke liye best hai.

Key: tagged = Agent Decides (Buggy)

Q55 **Medium** L9

TTS mein 1500 characters se lamba reply automatically kya hota hai aur limit kaise adjust karein?

A) Reply 2 separate voice notes mein split hoti hai

B) Auto-summarize ya skip ho jata hai — fix: WhatsApp pe `/tts limit 3000` run karo

C) Agent automatically text mein switch kar leta hai voice ki jagah

D) Gateway error throw karta hai aur retry karta hai

✅ **B) Auto-summarize/skip — /tts limit 3000**

TTS default char limit 1500 hai — longer replies summarize ya silently skip ho jaati hain. Limit badhao: WhatsApp mein type karo `/tts limit 3000`. Current status check: `/tts status`. Yeh limitation isliye hai ke bohot lamba audio note frustrating hota hai listeners ke liye.

Key: 1500 Char TTS Limit

Q56 **Easy** L10

Second agent create karne ke baad use WhatsApp se bind karne ka sahi command sequence kaun sa hai?

A) `openclaw agents connect helper-agent --whatsapp`

B) `openclaw agents add helper-agent` → `openclaw agents bind --agent helper-agent --bind whatsapp` → gateway restart

C) `openclaw config set agents.helper-agent.channel whatsapp`

D) Dashboard se manually drag and drop karo channel pe

✅ **B) agents add → agents bind → gateway restart**

3 steps: (1) `openclaw agents add helper-agent` — agent create. (2) `openclaw agents bind --agent helper-agent --bind whatsapp` — channel bind. (3) `openclaw gateway restart` — apply karo. Verify: `openclaw agents list --bindings`. Phir routing rules se specific senders ko route karo.

Key: Add → Bind → Restart

Q57 **Hard** L12

OpenClaw mein Session Lane aur Global Lane concurrency model ka kya matlab hai?

A) Session Lane fast hai, Global Lane slow hai — performance tiers hain

B) Session Lane = har customer ka isolated queue (parallel processing), Global Lane = sab agents share karte hain (bandwidth limit)

C) Global Lane customers ke liye, Session Lane admin ke liye reserved hai

D) Dono ek hi hain — alag marketing names hain

✅ **B) Session Lane = isolated per customer, Global Lane = shared bandwidth**

Session Lane: har customer ka apna isolated queue — multiple customers parallel chalte hain, ek doosre ko block nahi karte. Global Lane: sab agents share karte hain — ek shared bandwidth pool hai. Jab zyada subagents spawn hon to global lane pe congestion aa sakta hai. Isliye production mein model quality critical hai — cheap model zyada turns leta hai aur global lane bharta hai.

Key: Session Isolated, Global Shared

Q58 **Hard** L13

3-Tier Safety mein Tier 1 (Tool Profiles) aur Tier 3 (NemoClaw) ka fundamental enforcement difference kya hai?

A) Koi farq nahi — dono same cheez hain alag names ke saath

B) Tier 1 = in-process Node.js (same process jo enforce karta hai wahi use bhi karta hai — compromised ho sakta hai), Tier 3 = out-of-process kernel-level (alag isolated process — bypass possible nahi)

C) Tier 1 network security enforce karta hai, Tier 3 file system security

D) Tier 3 paid version mein hai, Tier 1 free mein hai

✅ **B) Tier 1 in-process (compromisable), Tier 3 out-of-process kernel-level**

Tier 1 (Tool Profiles): in-process Node.js — same process jo enforcement karta hai wahi bypass bhi kar sakta hai agar agent compromised ho. Tier 2 (Plugin Hooks): bhi in-process. Tier 3 (NemoClaw): out-of-process — Landlock (fs), seccomp (syscalls), netns (network) — OS kernel enforce karta hai, agent physically bypass nahi kar sakta chahe root access bhi mile.

Key: In-process vs Kernel-Level

Q59 **Hard** L15

Docker Compose mein API key ka security gap kya hai aur NemoClaw ise kaise fix karta hai?

A) Docker mein key unencrypted file mein hoti hai — NemoClaw encrypt karta hai

B) API key container environment variable mein hoti hai — attacker `printenv` run karwaye to exposed! NemoClaw mein Privacy Router alag pod mein key rakhta hai — agent ke environment mein koi key nahi hoti

C) Docker mein key automatically expire ho jati hai — NemoClaw auto-rotate karta hai

D) Docker Compose mein koi security gap nahi — NemoClaw sirf performance ke liye hai

✅ **B) Docker env var exposed — NemoClaw Privacy Router alag pod mein**

Docker Compose: `ANTHROPIC_API_KEY=sk-xxx` environment variable mein hoti hai. Attacker prompt inject kare → `printenv` → key exposed! NemoClaw fix: Agent endpoint =

`inference.local:8080` — koi API key nahi. Privacy Router alag network namespace ke pod mein real key rakhta hai → request receive → key add → provider forward. Agent ke paas root access bhi ho to bhi `printenv` mein koi key nahi milegi!

Key: Docker Gap vs Privacy Router Fix

Q60 **Hard** L15

NemoClaw ka Policy Engine kya enforce karta hai aur kaun se 3 mechanisms use karta hai?

A) Tool profiles enforce karta hai — same Node.js process mein

B) API key rotation schedule enforce karta hai

C) Landlock (file system access), seccomp (system calls), netns (network namespace) — kernel-level, alag pod mein out-of-process

D) Authentication aur user authorization enforce karta hai

✅ **C) Landlock + seccomp + netns — kernel-level**

Policy Engine (4th pod of NemoClaw): Landlock = file system access control (agent sirf allowed directories access kar sakta hai), seccomp = system call filtering (dangerous syscalls blocked at OS level), netns = network namespace isolation (sirf `inference.local` route allowed, direct internet connections blocked). Ye in-process checks nahi hain — OS kernel enforce karta hai alag pod se. Agent physically escape nahi kar sakta.

Key: Policy Engine = Kernel-Level (Landlock+seccomp+netns)

⭐ Section 8: Chapter 56 Synthesis (Q61)

Q61 **Hard** L1–L15

Chapter 56 ka overall journey kya demonstrate karta hai — Lesson 1 se Lesson 15 tak?

A) OpenClaw sirf hobbyists ke liye hai

B) AI employee banana ek simple 1-step process hai

C) Concept (6 dimensions) → Install → Capabilities (skills/MCP/voice/groups) → Proactive → Deploy → Security — chatbot se production AI Employee tak ka complete journey

D) Sirf coding ke liye useful hai

✅ **C) Concept → Install → Capabilities → Proactive → Deploy → Security**

L1: 6 dimensions samjhe. L2: Install kiya. L3-6: Agent loop, brain, memory, skills. L7-8: External tools + proactive. L9-11: Voice, multi-agent, Google. L12-13: Orchestration + security gates. L14-15: Production deployment + NemoClaw isolation. James ki story: "It works... and it remembers me. I told it my name ten minutes ago."

Key: Complete AI Employee Journey

🎉 Shabash Imran Hassan!

Chapter 56 ke tamam **15 Lessons ke 61 MCQs** complete kar liye!

AI Employee Moment → Install → Brain → Memory → Skills → MCP → Proactive → Voice → Multi-Agent → Deploy → Security

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